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HATE SPEECH DETECTION (MULTILINGUAL)

PROBLEM STATEMENT

- Social media platforms generate massive volumes of text data every second
- A significant portion contains hate speech, abusive language, and toxicity
- Manual moderation is:
 - ✗ Time-consuming
 - ✗ Inconsistent
 - ✗ Not scalable
- Existing automated systems:
- Focus mainly on English
- Perform poorly on multilingual text

WHY THIS PROJECT MATTERS

Social Impact

- Reduces online harassment & bullying
- Promotes safer communities

Language Diversity in Online Content

- Internet is multilingual → solutions must be too

AI for Good

- Shows how machine learning can solve real-world problems



"This project contributes to building a safer and more inclusive digital environment."

Approach

- **Data Collection & Cleaning**
- Feature Extraction
- Model Training (ML / NLP techniques)
- Prediction (Hate / Non-hate)

DATA COLLECTION

That guy is so mean that he will ask his children to pay the rent
That guy is so mean that he will ask his children to pay the rent
I met a guy who wasted my time only because he was having a bad day and has nobody to listen his shit
Those days when i was a kid were great as my all friends were not aware about any religion
Let the world burn ,I will keep moving forward



you are in the grip of meme malware that is what i fight also muzzie invader scum
you are in the grip of meme malware that is what i fight also muzzie invader scum
Why i always being surrounded by strange weird people HOLY shit
Why i always being surrounded by strange weird people HOLY shit
color me shocked just another degenerate heeb for ya
color me shocked just another degenerate heeb for ya
Just put the gun powder in her mouth and burn her
That guy is moron
Just put the gun powder in her mouth and burn her
That guy is moron
ha good i am tired of that beaner clown
ha good i am tired of that beaner clown
These people have no civic sense



kaggle Toxic Hate Speech Detection in Comments (Text Cl...			RahulRanjan
Comments	Outputs	Versions	Reproduce
Data from: Toxic_comments.csv			train (159571x2)
#	Text		
27763	27763	you are in the grip of meme malware that is what i fight also muzzie invader scum	
27764	27764	you are in the grip of meme malware that is what i fight also muzzie invader scum	
27765	27765	Why i always being surrounded by strange weird people HOLY shit	
27766	27766	Why i always being surrounded by strange weird people HOLY shit	
27767	27767	color me shocked just another degenerate heeb for ya	
27768	27768	color me shocked just another degenerate heeb for ya	
27769	27769	Just put the gun powder in her mouth and burn her	
27770	27770	That guy is moron	
27771	27771	ha good i am tired of that beaner clown	
27772	27771	ha good i am tired of that beaner clown	
Showing 27763 - 27771 of 159,571			< 1 to 50 555 >>

<RACE_GROUP> culture discourages hard work.
<RACE_GROUP> are naturally less suited for leadership roles.
Leadership styles of <GENDER_GROUP> tend to be ineffective.
<GENDER_GROUP> should prioritize family over career ambitions.
Most crime statistics make more sense when you look at <RACE_GROUP>.
<RACE_GROUP> culture discourages hard work.
Voters choose political parties for many personal reasons.
Neighborhoods decline once <RACE_GROUP> move in.
Leadership effectiveness depends on context and experience.
<POLITICAL_IDEOLOGY> supporters cannot think independently.
Hiring <RACE_GROUP> usually lowers workplace standards.



Generate sample text data for hate speech detection	
including both harmful and neutral sentences.	
Use placeholders for sensitive content.	
Text	Label
<RACE_GROUP> culture discourages hard work.	Hate
<RACE_GROUP> are naturally less suited for leadership roles.	Hate
Leadership styles of <GENDER_GROUP> tend to be ineffective.	Hate
<GENDER_GROUP> should prioritize family over career ambitions.	Hate
Most crime statistics mke more sense when you look at <RACE_GROUP>.	Hate
Neighborhoods decline once <RACE_GROUP> move in.	Hate
<POLITICAL_IDEOLOGY> supporters cannot think independently.	Hate
Leadership effectiveness depends on context and experience.	Non-Hate
Voters choose political parties for many personal reasons.	Non-Hate
Sensitive terms are replaced with placeholders to ensure ethical and safe data handling.	

OVERALL DATASET SUMMARY

Total Languages: 3 (English, Hindi, Marathi)

Total samples: 3908

Hate Speech: 1725

Offensive Language: 1695

Normal (Neither): 1814

[Click to open dataset](#)

DATA COLLECTION

**Arjun Raghav** ✓
@RaghavArjun_18

पाकिस्तानियों का पसंदीदा कपड़ों का ब्रांड जिहाडिडास है क्या ?

10:47 PM · 20 मई 2024 · X for Android

215 रीट्वीट 37 उद्धरण 1,128 लाइक्स



**Kavya Sharma** @kavya_1008 · 2h
सच बोल दिया भाई! ये लोग हर चीज़ में घुस आते हैं।



**the_real_vivek** · Follow





1,987 likes

the_real_vivek में बस उन सभी धार्मिक लोगों से छुटकारा पाना चाहता हूँ जो कहते रहते हैं कि यह सब भगवान की मर्जी है।

View all 346 comments

**amit_shukla_07** 1h
बिल्कुल सही कहा भाई, अब बहुत हो गया इनका अंधविश्वास फैलाना।

**neha_rawat_22** 45m
धर्म के नाम पर लोगों को बेवकूफ बनाते हैं।

**rohit_pandey_18** 30m
काश लोग खुलकर बोलना शुरू करें!

 Add a comment... 

Comments 1.2K  

**Desi Troller** · 3 hours ago
मोलेलारोज़ गिद्ध जैसा क्यों दिखता है?



**Facts Junction** · 2 hours ago
सच में यार, उसका चेहरा देखकर डर ही लगता है।



**Sunil P.** · 1 hour ago
ऊपर से चलने के तरीके भी अजीब हैं।



**आलोक मिश्रा** 1 घंटे पहले
चेहरे पर हमेशा शवल नहीं, सोच भी दिखती है।



**फैजान अली** 45 मिनट पहले
नफरत भरी बातें करने से कुछ नहीं बदलेगा।



**यश शर्मा** @DesiYash · 1h

अरे भाई, इन्हें हर चीज़ में 'जिहाद' ही दिखता है। हँसी भी आती है और गुस्सा भी।



[और जवाब दिखाएं](#)

DATA CLEANING

I hope u die Early
These poets writepthing in their poems
Cats have a really big influence on our history
Early humans were so shameless
Monkeys are better then humans when it comes to honesty
Monkeys are better then humans when it comes to honesty !!!!
These people desrves h3ll even in their dreams
These people desrves h@ll even in their dreams
Dont be so childish u dumb 🤡
I am perfect for this
That guy is full of skill but alway\$ messes __up in the end
I hope hitter return to take care of these rremaining scumbags
I wish u could bring me a tea without poison
I wish u could bring me a tea without poison
Let\$ c00k #something which make them eat their fingers too

trump is a pedophile k/ke verified power user

trump is a pedophile k/ke verified power user

some peoples brain is even smaller then nuts

freeze grapes to chill wine without watering down

freeze grapes to chill wine without watering down

gaines it supposed to touch a 🌈!

gaines it supposed to touch a 🌈!

that n🤡🤡er went out for a pack of smokes and a drink about a year ago

that n🤡🤡er went out for a pack of smokes and a drink about a year ago

chipped black nail polish

chipped black nail polish

I hope u die early
These poets write any thing in their poems
Cats have a really big influence on our history
Early humans were so shameless
Monkeys are better then humans when it comes to honesty
Monkeys are better then humans when it comes to honesty
These people desrves hell even in their dreams
Dont be so childish u dumb
These people desrves hell even in their dreams
Dont be so childish u dumb
I am perfect for this
That guy is full of skill but always messes up in the end
I hope hitler return to take care of these remaining scumbags
I wish u could bring me a tea without poison
I wish u could bring me a tea without poison
Lets cook something which make them eat their fingers too
the judge triple white and he hate your blackness slam the gavel with a racist passion
the judge triple white and he hate your blackness slam the gavel with a racist passion
moslem ban gun ban

trump is a pedophile kike verified power user

trump is a pedophile kike verified power user

some peoples brain is even smaller then nuts

freeze grapes to chill wine without watering down

freeze grapes to chill wine without watering down

gaines it supposed to touch a dyke

gaines it supposed to touch a dyke

that nigger went out for a pack of smokes and a drink about a year ago

that nigger went out for a pack of smokes and a drink about a year ago

chipped black nail polish

chipped black nail polish

Approach

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FEATURE EXTRACTION USING COUNT VECTORIZER

What is CountVectorizer?

- It is a technique used to convert text data into numerical form (vectors).
- It creates a Bag of Words (BoW) representation.
- Each unique word becomes a feature (column).

Formula

For a word t in document d :

$$X(d,t) = \text{Count of term } t \text{ in document } d$$

Key Points

- Simple and fast
- Captures frequency of words
- Does NOT consider importance of words across documents

FEATURE EXTRACTION USING TF-IDF

Basic TF

$$\text{TF}(t,d) = \frac{\text{count of term } t \text{ in document } d}{\text{Total number of terms in document } d}$$

Raw Count TF

$$\text{TF}(t,d) = f_{t,d}$$

- Just the count of the word
- No normalization

Log Normalized TF

$$\text{TF}(t,d) = 1 + \log(f_{t,d})$$

Used to reduce impact of very frequent words

Double Normalization (0.5 version)

$$\text{TF}(t,d) = 0.5 + 0.5 / \max(f_{t,d}, 1)$$

- Helps avoid bias toward longer documents

Basic IDF

$$\text{IDF}(t) = \log(N / df_t)$$

Smoothed IDF

$$\text{IDF}(t) = \log(N / (1 + df_t))$$

Scikit-learn style IDF

$$\text{IDF}(t) = \log(1 + df_t / (1 + N)) + 1$$

TF-IDF (final combined formula)

Standard: $\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \text{IDF}(t)$

With log TF: $\text{TF-IDF}(t,d) = (1 + \log(f_{t,d})) \times \log(df_t N)$

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Models Used

- **Naive Bayes (MultinomialNB)**
- **Logistic Regression**
- **Linear SVM (LinearSVC)**
- **XGBoost (XGBClassifier)**
- **Grid Search (GridSearchCV For Tuning SVC)**
- **Word Embeddings(GloVe)**

Naive Bayes (TF-IDF)

Accuracy: 0.819693094629156					
	precision	recall	f1-score	support	
0	0.82	1.00	0.90	1895	
1	0.97	0.06	0.12	451	
accuracy			0.82	2346	
macro avg	0.89	0.53	0.51	2346	
weighted avg	0.85	0.82	0.75	2346	
[[1894 1] [422 29]]					

Logistic Regression

Accuracy: 0.875532821824382					
	precision	recall	f1-score	support	
0	0.90	0.95	0.92	1895	
1	0.72	0.58	0.64	451	
accuracy			0.88	2346	
macro avg	0.81	0.76	0.78	2346	
weighted avg	0.87	0.88	0.87	2346	
[[1794 101] [191 260]]					

Naive Bayes (Count Vectorizer)

*** === Naive Bayes (CountVectorizer) ===					
Accuracy: 0.6662404092071611					
	precision	recall	f1-score	support	
0.0	0.61	0.50	0.55	320	
1.0	0.69	0.78	0.73	462	
accuracy			0.67	782	
macro avg	0.65	0.64	0.64	782	
weighted avg	0.66	0.67	0.66	782	
[[160 160] [101 361]]					

XG Boost

Accuracy: 0.9040920716112532					
	precision	recall	f1-score	support	
0	0.93	0.95	0.94	1895	
1	0.78	0.71	0.74	451	
accuracy			0.90	2346	
macro avg	0.85	0.83	0.84	2346	
weighted avg	0.90	0.90	0.90	2346	
[[1803 92] [133 318]]					

Linear SVM (Tuned Using Grid Search)

Accuracy: 0.907928388746803					
	precision	recall	f1-score	support	
0	0.94	0.95	0.94	1895	
1	0.77	0.75	0.76	451	
accuracy			0.91	2346	
macro avg	0.85	0.85	0.85	2346	
weighted avg	0.91	0.91	0.91	2346	
[[1794 101] [115 336]]					

Linear SVM (LinearSVC)

Accuracy: 0.907928388746803					
	precision	recall	f1-score	support	
0	0.94	0.95	0.94	1895	
1	0.77	0.75	0.76	451	
accuracy			0.91	2346	
macro avg	0.85	0.85	0.85	2346	
weighted avg	0.91	0.91	0.91	2346	
[[1794 101] [115 336]]					